AIORI-3 Hackathon · PS-005 · Cyber Security

HORA Q

Explainable BGP Origin and Subprefix Anomaly Monitor



Poisoning-Aware Baseline Protection · Evidence Intelligence · Post-Quantum Verification

Team code: A3-PS005-TC416

Sankardev VS (Team Lead) · Adarsh A · Dr. Jyothy Joseph (Faculty Mentor)
Al-Ameen College

Abstract and Problem Description

Abstract

HORA Q is a prototype for detecting selected BGP origin-AS changes and new more-specific prefixes. It correlates observations with history, provenance and available RPKI context to give explainable findings. A cautious baseline policy keeps new or conflicting observations from being trusted until defined evidence criteria are met. A Post-Quantum Evidence Vault signs investigation records with ML-DSA for independent verification. A replay laboratory compares the workflow with a simple reference detector.

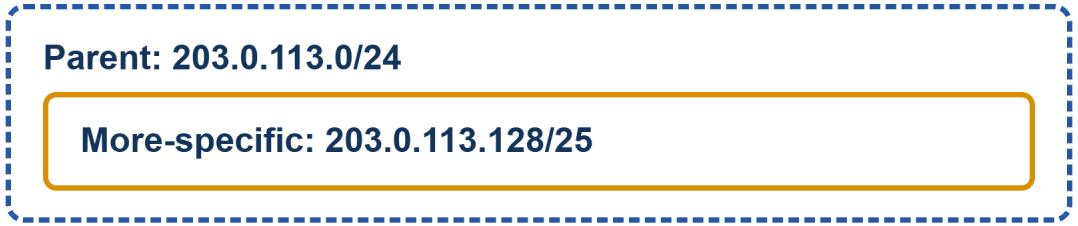
Problem Description

- Origin changes and more-specifics may be malicious, accidental, planned or benign
- Basic alerts lack history and evidence provenance
- Unverified observations can undermine a local baseline
- Incident records need evidence, rule versions and timestamps
- Signatures must verify independently without proving an observation true

Problem to solve: how can a BGP monitor produce explainable anomaly findings, protect its baseline from unverified observations and preserve independently verifiable incident evidence?

Background

BGP lets autonomous systems exchange reachability for IP prefixes. A changed origin AS or a new more-specific prefix can indicate an anomaly, but neither alone proves malicious intent. RPKI origin validation adds context on whether an origin AS is authorized under available ROA data [2, 3]; it does not establish full-route safety or intent. HORA Q treats observations as evidence with provenance, timestamps and limits, keeps a versioned baseline, and leaves conflicting or insufficient evidence visible.



Documentation example, not a real incident.

Origin-AS change

Observed origin differs from the reference state

More-specific prefix

Longer prefix covering part of its parent

Candidate observation

New evidence not yet promoted to the baseline

Evidence provenance

Source, timestamp and rule versions behind it

Literature and RFC Review

Standards

RFC 4271 defines BGP-4 [1]. RFC 6811 (updated by RFCs 8481 and 8893 [9, 10]) and RFC 9582 cover prefix-origin validation and ROAs [2, 3].

Existing Tools

BGPalerter supports unexpected-origin and more-specific alerts [4]. ARTEMIS is an established hijack-detection system [5]. HORA Q claims no novelty for basic detection.

Research Risk

Research reports that crafted public observations can mislead certain ML-based hijack detectors [6]. HORA Q studies baseline contamination in controlled scenarios; it does not claim to solve all poisoning attacks.

HORA Q Contribution

An integrated, testable workflow: cautious baseline promotion, provenance-aware explanations, explicit uncertainty, ML-DSA-signed records [7] and independent verification, evaluated against a reference detector.

Proposed Solution: Seven-Stage Workflow

1. **Ingest & normalize**
2. **Maintain history**
3. **Evaluate candidates**
4. **Detect anomalies**
5. **Correlate & explain**
6. **Sign & verify**
7. **Review & evaluate**

Baseline Promotion Policy

Evaluate source provenance, freshness, historical consistency, corroboration and validation context. Record each decision and rationale; keep observations as candidates when evidence is insufficient or conflicting.

Post-Quantum Evidence Vault

Stores canonical records, evidence hashes, algorithm, key ID and verification metadata. ML-DSA protects record integrity relative to the key; it does not prove the observation is true or an attack occurred.

Safety boundary: HORA Q does not automatically change network routes.

Reference Detector vs. Enhanced HORA Q

Dimension	Naïve experimental reference detector	Proposed HORA Q
Detection	Origin-change and subprefix rules	Same rules plus evidence correlation
Baseline	Incorporates observations immediately	Explicit candidate and promotion criteria
Explanation	Alert and changed route fields	Rule, history, provenance, evidence gaps
Evidence integrity	Basic finding record	ML-DSA-signed record, independent verifier
Evaluation	Detection results	Detection, baseline, evidence, crypto metrics

Detection
precision, recall, F1, false-positive rate

Baseline protection
contamination, legitimate-change withholding, promotion delay

Evidence quality
field completeness; conflicting or missing evidence handled

Crypto overhead
sign/verify latency, signature size, tamper tests

Evaluation status: planned measurements; no improvement is claimed until experiments are complete. The reference detector is an experimental comparison policy, not a claim about all monitors.

Solution Architecture

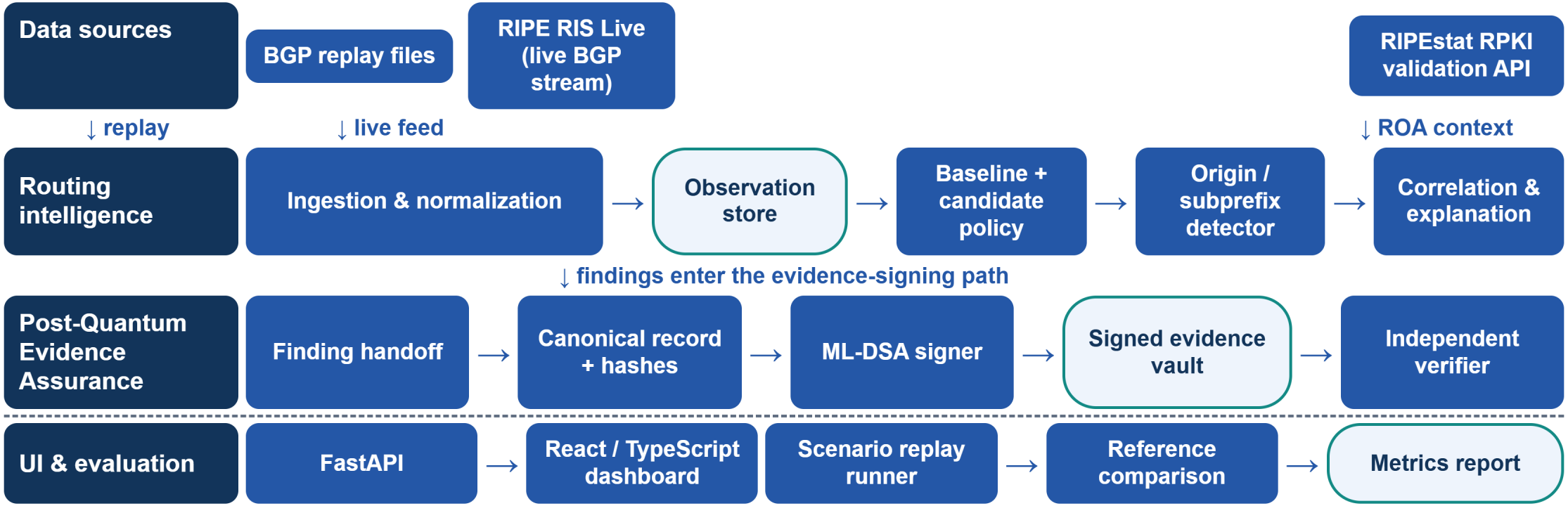


Figure 1. HORA Q component architecture. Routing intelligence generates explainable findings; the evidence layer signs and independently verifies investigation records. Teal = data store; Replay mode stays operational if the live stream or RPKI service is unreachable. The verifier consumes a saved record and public key; the dashboard and benchmark path stays outside signing.

Data Flow Diagram

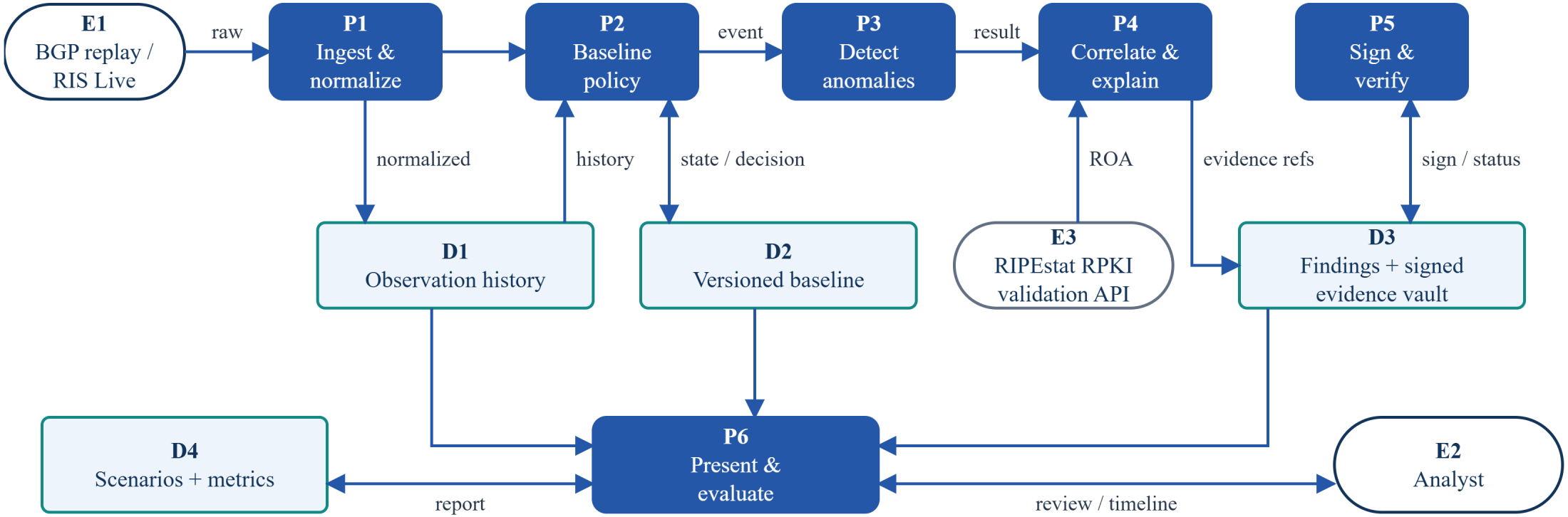


Figure 2. HORA Q data flow: observations become reviewed findings and signed evidence; scenario results are retained for reproducible evaluation. Blue = process, teal = data store, white = external entity.

Deployment Timeline

12–14 Oct Setup | **16 Oct** Level 2 | **16–19 Oct** Sprint 1 | **20–24 Oct** Sprint 2 | **25–27 Oct** Sprint 3 | **28–30 Oct** Sprint 4 | **By 31 Oct** Steps 8–10

Stage	Milestone	Key deliverable
Setup	Submission readiness	Proposal PDF, repository, intro video
Level 2	Team announcement	Official announcement; mentoring starts
Sprint 1	Core detection	Replay format; origin/subprefix tests
Sprint 2	Baseline protection	Candidate policy, correlation, explanations
Sprint 3	Evidence assurance	ML-DSA signing, verifier, tamper tests
Sprint 4	Integration and evaluation	Dashboard, scenario suite, comparisons
Steps 8–10	Final deliverables	Source, documentation, working demo

Methodology: Plan → Implement → Test → Integrate → Benchmark → Document. Dates are planning targets; check official AIORI-3 announcements [8].

Bibliography

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https://www.rfc-editor.org/rfc/rfc8481

[10] “RPKI Origin Validation for BGP Export,” RFC 8893, IETF, 2020.
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Thank You

HORA Q

Explainable evidence. Verifiable routing investigations.

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github.com/vssankardev-lang/A3-PS005-TC416